SentinelAI

Adaptive Fraud Intelligence Platform

CyberShield Hackathon | Problem Statement 2 | Bank of India

PROFESSIONAL IDEATION DOCUMENT — VERSION 2.0

AI/ML Classification of Suspicious Mule Accounts

Fraud Genome Engine | **Anomaly Detection** | **Explainable AI (XAI)** | **Fraud Evolution Intel**

CONFIDENTIAL — FOR HACKATHON EVALUATION ONLY

2025

Feature 3924 Target | 3924-Dim Dataset | India Deployment Ready

Problem Analysis & Executive Summary

PROBLEM STATEMENT 2 — ANALYSIS

Banks face a growing epidemic of mule accounts — bank accounts used to receive, layer, and transfer stolen money before it disappears. These accounts are the key infrastructure of cyber-enabled financial fraud.

The fraud chain:

- Criminal steals victim's money via phishing/scam
- Stolen funds sent to mule account at a bank
- Mule account layers funds across 4-6 accounts
- Money withdrawn as cash or crypto — gone

Why current systems fail:

- Rule-based: rigid thresholds, easily bypassed
- No real-time scoring — money moves in seconds
- No explanation — investigators fly blind
- Cannot adapt as criminal methods evolve
- Feature 3924 imbalance — <0.3% positive cases

Target: Feature 3924

Mule Account (1) vs Legitimate (0)

18 known indicator features provided

SENTINELAI — EXECUTIVE SUMMARY

SentinelAI is an Adaptive Fraud Intelligence Platform that goes beyond binary classification to deliver behavioural intelligence, unseen-pattern detection, and explainable risk decisions.

Five core differentiators:

- **Fraud Genome Engine**
  Groups mule accounts into behavioural archetypes using clustering — tells investigators WHAT type of fraud
- **Anomaly Detection**
  Isolation Forest + Autoencoder catches new fraud patterns that the supervised model has never seen before
- **Explainable AI (XAI)**
  SHAP values explain every risk score — which features drove the flag and by how much
- **Fraud Evolution**
  Monitors cluster drift over time to detect emerging criminal strategies before losses spike
- **India-First Design**
  Built for DPDP Act, RBI validation, FIU-IND STR output, and Hindi XAI from day one

PROBLEM STATEMENT 2 — REQUIREMENT ALIGNMENT

Feature engineering on transaction data	5 feature families + 18 known indicator analysis
ML models for behavioural pattern learning	XGBoost + RF ensemble with SMOTE for imbalance
Anomaly detection for new fraud patterns	Isolation Forest + Autoencoder unsupervised layer
Predictive risk scoring	Unified 0-100 score combining classifier + anomaly
Intelligent alert generation	SHAP XAI + FIU-IND STR + Investigator action guide

INPUT LAYER

- **CBS Transaction Data** — Core banking real-time event stream (Kafka)
- **Account Metadata** — Customer profile, KYC, account age
- **UPI Payment Signals** — NPCI VPA data, collect requests

FEATURE ENGINEER

- **Feature Engineering** — 5 families from 3,924 dimensions extracted
- **Data Preprocessing** — SMOTE oversampling + normalization pipeline
- **Quality Validation** — Missing value impute, outlier detection

INTEL LAYER

- **ML Ensemble Classifier** — XGBoost + Random Forest — supervised
- **Anomaly Detector** — Isolation Forest + Autoencoder — unsup.
- **Fraud Genome Engine** — K-Means + DBSCAN clustering layer

OUTPUT LAYER

- **Unified Risk Score** — 0-100 score combining classifier + anomaly
- **XAI Explanation** — SHAP values + Hindi investigator report
- **Alert + FIU-IND STR** — Automated STR in RBI XML format

REAL-TIME PROCESSING PIPELINE — END-TO-END < 100ms

- **Transaction Event** — Kafka consumer — t=0ms
- **Feature Extraction** — Redis feature store lookup — <10ms
- **Fast Model Scoring** — Tier-1 gradient boosted tree — <50ms
- **Genome Matching** — Nearest centroid assignment — <30ms
- **X Gen** — SHAP + template rendering — <10ms

Total
< 100ms
end-to-end

THE 18 KNOWN FRAUD INDICATOR FEATURES — CATEGORY ANALYSIS

NOTE: Feature names are anonymised (F115, F321...). EDA will reveal exact semantics. Categories above are inferred from feature index clustering and domain knowledge.

Transaction Activity	Network Behaviour	Value Patterns	Account Lifecycle	Risk Signals
F115	F670	F2582	F3043	F3891

5 ENGINEERED FEATURE FAMILIES — DERIVED FROM 3,924 RAW FEATURES

Transaction Velocity

- Tx count per 7/30/90 day windows
- Transactions per hour (burst metric)
- Average inter-transaction interval
- Velocity change ratio (recent vs history)

Amount Distribution

- Mean, std dev, skew of tx amounts
- Ratio of round-number transactions
- Peer-group amount deviation score
- Max/min ratio in rolling 30-day window

Network Patterns

- Unique counterparty count (7/30d)
- New beneficiary ratio per window
- Counterparty reuse concentration
- Fan-out vs fan-in ratio (receive/send)

Temporal Signals

- Night-time transaction ratio (11pm-5am)
- Weekend vs weekday activity ratio
- Transaction burst window concentration
- Dormancy period length before activity

Account Lifecycle

- Account age at time of flag
- Days since KYC vs days active ratio
- Balance trajectory slope (7/30d)
- Credit/debit ratio in rolling windows

CLASS IMBALANCE HANDLING STRATEGY — FRAUD < 0.3% OF ACCOUNTS

SMOTE Oversampling

Synthetic minority oversampling — generate synthetic fraud examples to achieve 10:1 legitimate-to-fraud ratio in training

Focal Loss Function

Modified cross-entropy that down-weights easy negative examples — model focuses training effort on hard fraud cases

Cost-Sensitive Learning

Misclassifying fraud = 100x penalty over false alarm — model learns that missing fraud is catastrophically expensive

Evaluation Protocol

NEVER use accuracy. Primary: F1-score + AUC-PR. Secondary: AUC-ROC. Test on stratified hold-out preserving 0.3% ratio

ML Model Architecture — Ensemble Design

PRIMARY MODEL — XGBOOST CLASSIFIER

Why XGBoost for mule detection?

Gradient boosting handles non-linear fraud patterns naturally, is robust to missing features (common in banking data), and provides built-in feature importance.

Configuration:

n_estimators	500 trees
max_depth	6 (prevents overfit)
learning_rate	0.05 (conservative)
scale_pos_weight	333 (333:1 imbalance)
eval_metric	aucpr (PR curve)
colsample_bytree	0.8 (feature sampling)
subsample	0.8 (row sampling)

Expected output: probability score 0.0 to 1.0

ENSEMBLE BASE — RANDOM FOREST

Why Random Forest as ensemble base?

RF provides diverse base learners and excellent feature importance ranking via Gini impurity — helps validate which of the 18 known features matter most.

Configuration:

n_estimators	300 trees
max_depth	12 (deeper allowed)
min_samples_leaf	20 (reduce overfit)
class_weight	balanced_subsample
max_features	sqrt (standard RF)
bootstrap	True (bagging enabled)
criterion	gini + entropy blend

Expected output: probability score 0.0 to 1.0

ENSEMBLE COMBINATION STRATEGY

Final Risk Score = (0.6 x XGBoost probability) + (0.4 x Random Forest probability) + Anomaly bonus

- **XGBoost weight: 60%** — Higher weight — generally more accurate on structured tabular data with class imbalance
- **RF weight: 40%** — Lower weight — provides diversity and reduces variance from XGBoost overconfidence
- **Anomaly bonus: +10pts** — If Isolation Forest flags the account as an outlier, +10 pts added to the raw score
- **Score normalization** — Final weighted sum rescaled to 0-100 range for investigator-friendly output

MODEL TRAINING & VALIDATION STRATEGY

- **Data Split** — 70% train / 15% validation / 15% test — all stratified to preserve 0.3% fraud ratio
- **Cross-validation** — 5-fold stratified CV on training set — ensures robust performance estimate across folds
- **Hyperparameter tuning** — Bayesian optimisation (Optuna) on validation AUC-PR — 100 trial budget
- **Retraining trigger** — Automatic quarterly retrain + on-demand when Evidently AI detects >5% drift
- **Model versioning** — MLflow tracks all experiments — every model version reproducible with full lineage

KEY INNOVATION 1 — FRAUD GENOME ENGINE: BEHAVIOURAL ARCHETYPES

Just as human DNA defines biological characteristics, fraud leaves distinctive behavioural signatures. The Fraud Genome Engine groups confirmed mule accounts into archetypes using K-Means + DBSCAN clustering — enabling investigators to understand not just THAT fraud occurred, but WHAT type.

Cluster A

High-Velocity Micro-Transfer

- Hundreds of small txns (avg. <500) in rapid bursts
- Funds dispersed to 20+ new beneficiaries in 48hrs
- Account typically 1-3 months old
- High night-time activity ratio (>40%)
- Typical fraud: smurfing / layering

Cluster B

Dormant-Burst Pattern

- Account inactive 6-12 months, then sudden activity
- Large single transfer followed by rapid dispersal
- Original transaction pattern completely changes
- New beneficiaries = 90%+ of post-dormancy txns
- Typical fraud: account takeover / compromised

Cluster C

Collector Account

- Receives funds from many different sources
- Inbound-only or heavily inbound-skewed
- High counterparty diversity on receive side
- Funds aggregate then move in one large transfer
- Typical fraud: aggregation node in mule chain

Cluster D

Synthetic Normal Pattern

- Engineered to look like regular consumer account
- Anomaly detector flags it despite normal surface
- Subtle statistical deviations in timing patterns
- Peer-group deviation score unusually high
- Typical fraud: AI-designed evasion account

The ML classifier catches known fraud patterns. The Anomaly Detector catches UNKNOWN ones — brand-new fraud methods the model has never seen. Two models work in parallel.

KEY INNOVATION 2 — ANOMALY DETECTION: CATCHING THE UNKNOWN

Isolation Forest

- Randomly partitions features — anomalies isolated faster
- contamination=0.03 to 0.05 (expected fraud rate)
- 100 estimators for stability
- Works on raw feature vectors — no labels needed
- Generates anomaly score 0-1 (higher = more unusual)

Autoencoder Neural Network

- Encoder compresses account features to latent space
- Decoder reconstructs original feature vector
- High reconstruction error = account behaves unusually
- Trained ONLY on legitimate account data
- Threshold set at 95th percentile of reconstruction error

Combined Anomaly Score

- Average of Isolation Forest + Autoencoder scores
- Account flagged if combined score > 0.7 threshold
- Anomaly score added as bonus to final risk score
- Catches fraud Cluster D (synthetic normal pattern)
- Reviewed monthly — threshold tuned as fraud evolves

Explainable AI (XAI) & Investigator Workflow

SHAP-BASED EXPLAINABLE AI — WHY EVERY FLAG IS JUSTIFIED

SHAP (SHapley Additive exPlanations) calculates the exact contribution of every feature to each individual prediction. Unlike global feature importance, SHAP gives a per-account, per-decision explanation.

- **Per-prediction SHAP** — Each flagged account gets its own SHAP waterfall — not the same explanation for all
- **Feature direction** — SHAP shows both the magnitude (+/-) and direction (raises vs lowers risk score)
- **Hindi output** — XAI explanation template translated into Hindi for tier-2/3 bank investigators

EXAMPLE INVESTIGATOR OUTPUT — WHAT THE SYSTEM ACTUALLY SHOWS

FLAGGED ACCOUNT: ACC-7291847

Flagged: 14 Jun 2025, 03:42 AM | Branch: Mumbai-Central
Review required within: 4 hours

RISK SCORE	FRAUD GENOME	ANOMALY SCORE
87 / 100	Cluster B — Dormant-Burst Pattern	74.2 / 100 (Isolation Forest)

WHY THIS ACCOUNT WAS FLAGGED — TOP SHAP VALUE DRIVERS:

F670	**New Beneficiary Ratio**	94% of transfers went to accounts never transacted with before	+0.38
F321	**Avg. Transfer Amount**	23.4x higher than peer-group average for similar account type	+0.31
F2082	**Transaction Burst Metric**	47 transactions completed within a 2-hour window at 3 AM	+0.28
F527	**Account Dormancy Flag**	Account was inactive for 8 months before sudden activation	+0.22

RECOMMENDED ACTION: Enhanced 30-day monitoring + Call customer on registered mobile + Pre-generate STR for FIU-IND

12 other accounts share this fraud genome cluster — consider joint investigation

INVESTIGATOR ACTION FRAMEWORK — DECISION MATRIX BY RISK BAND

90 – 100	CRITICAL	Immediate account hold + Auto-generate FIU-IND STR + Escalate to bank CISO within 1 hour
75 – 89	HIGH	Enhanced 30-day monitoring + Call customer on registered mobile + Pre-generate STR draft
60 – 74	MEDIUM	Add to watchlist + Flag next transaction for manual review + Inform branch manager
40 – 59	LOW	Log for trend analysis + Re-evaluate in 14 days + No customer-facing action
0 – 39	NORMAL	No action — account cleared. Log SHAP explanation for audit trail and model feedback

Evaluation Framework & Expected Performance

THE ACCURACY TRAP — WHY WE NEVER REPORT ACCURACY

With <0.3% fraud rate, a model that labels EVERY account as legitimate achieves 99.7% accuracy. This model catches zero fraud. Accuracy is meaningless for imbalanced fraud detection. Our evaluation uses precision-recall metrics that explicitly account for class imbalance.

PRIMARY EVALUATION METRICS — DEFINITION AND TARGETS

~0.85

F1-Score

Harmonic mean of Precision and Recall. Primary metric — balances false positives (investigator wasted time) against false negatives (fraud

~0.91

AUC-PR

Area Under Precision-Recall Curve. Best metric for severe class imbalance. Measures how well we detect fraud at all classification thresholds.

~0.94

AUC-ROC

Area Under ROC Curve. Measures discrimination between fraud and legitimate at all thresholds. Used for model comparison. Target:

< 8%

False Pos. Rate

Fraction of legitimate accounts wrongly flagged. Critical — drives investigator workload and customer complaints. Target: <8% at operating

CROSS-VALIDATION + TESTING STRATEGY

- **Stratified 5-fold CV** — Training set split into 5 folds, each fold preserves 0.3% fraud ratio. Hyperparameters tuned on mean validation
- **Hold-out test set** — 15% of data held out — never seen during training or CV. Final performance reported on this set only.
- **Temporal validation** — If timestamps available, also validate on last 20% chronologically — tests whether model degrades on newer frau
- **Adversarial test** — Manually construct 50 accounts designed to look normal but matching known fraud patterns — tests model robustnes

EXPECTED PERFORMANCE — SENTINELAI vs BASELINE COMPARISON

Metric	Rule-Based Baseline	XGBoost Alone	SentinelAI Ensemble	SentinelAI + Anomaly
F1-Score	0.41	0.78	0.85	0.87
AUC-PR	0.38	0.87	0.91	0.93
AUC-ROC	0.61	0.91	0.94	0.95
False positive %	22%	11%	7.5%	7.2%
New fraud caught	No	No	No	Yes
Explanation	None	None	SHAP	SHAP

Implementation Roadmap, Tech Stack & Compliance

COMPLETE TECHNOLOGY STACK

Data & Processing	ML & Modelling	Explainability & Monitoring	Infrastructure & Deployment	Compliance & Security
Python 3.11	XGBoost 2.0	SHAP library	FastAPI (scoring API)	Immutable audit log
pandas / numpy	scikit-learn RF	Evidently AI (drift)	Docker / Kubernetes	AES-256 encryption
scikit-learn	TensorFlow (Autoencoder)	Plotly (dashboards)	PostgreSQL (audit logs)	RBAC access control
Apache Kafka	imbalanced-learn (SMOTE)	Sarvam AI (Hindi NLP)	AWS/Azure (India region)	CERT-In audit trail
Redis (feature store)	Optuna (HPO)	Prometheus (metrics)	HashiCorp Vault (secrets)	DPDP consent layer
Apache Spark	MLflow (versioning)	Grafana (monitoring)	NGINX (load balancer)	FIU-IND XML reporter

4-PHASE IMPLEMENTATION ROADMAP

Phase 1 — 0-3 months — Foundation

- EDA on Bank of India dataset
- Feature engineering pipeline
- XGBoost + RF training & tuning
- SHAP integration + Hindi XAI
- RBI model card documentation

Phase 2 — 3-6 months — Intelligence

- Anomaly Detection layer (IF + AE)
- Fraud Genome clustering engine
- Fraud Evolution monitoring
- Investigator dashboard (FastAPI)
- FIU-IND STR auto-generator

Phase 3 — 6-9 months — Production

- Kafka real-time scoring pipeline
- CBS (Finacle/BaNCS) integration
- CERT-In security audit
- RBI regulatory sandbox pilot
- DPDP consent management layer

Phase 4 — 9-12 months — Scale

- Multi-bank federated deployment
- Drift monitoring + auto-retrain
- Hindi + regional language XAI
- Commercial SLA framework
- Full production go-live at BOI

INDIA REGULATORY COMPLIANCE CHECKLIST

- ok **DPDP Act 2023** — Consent management module built in. Purpose limitation enforced.
- ok **RBI Model Risk** — Model cards + governance docs + validation-ready design from day one.
- ok **CERT-In Cybersec** — Pre-certification security audit pathway with empanelled auditor.
- ok **FIU-IND STR** — Auto-generates Suspicious Transaction Reports in prescribed XML format.
- ok **CBS Integration** — Finacle + BaNCS plugin partnership distribution strategy defined.
- ok **Hindi XAI** — Explanation engine generates Hindi output. Regional languages roadmapped.

Business Case & Conclusion

QUANTIFIED IMPACT FOR BANK OF INDIA — KEY PERFORMANCE INDICATORS

40-60%	85%+	< 8%	< 100ms	100%
Reduction in manual review time	Fraud detection precision	False positive rate	Real-time risk scoring	Decision audit trail

COST-BENEFIT ANALYSIS — SIMPLIFIED

WITHOUT SENTINELAI

- Rule-based system: misses 60%+ of mule accounts
- Average fraud loss per mule account: Rs 2-8 lakh
- Investigator time: 2-3 hrs per manual review
- Regulatory risk: inadequate STR filing
- Reputational damage from undetected fraud chains

WITH SENTINELAI

- Detects 85%+ of mule accounts vs 40% rule-based
- Every flag comes with an explanation — faster action
- Investigator time: 20 min per auto-explained flag
- FIU-IND STR auto-generated — compliance protected
- Fraud evolution tracking prevents pattern repeat

SENTINELAI vs TRADITIONAL RULE-BASED SYSTEMS — QUICK COMPARISON

Capability	Rule-Based	SentinelAI
New fraud pattern detection	No — blindspot	**Yes — Anomaly layer**
Explanation for each flag	No — black box	**Yes — SHAP + Hindi**
Fraud behaviour clustering	No — account-by-account	**Yes — Fraud Genome**
Adaptive to evolving fraud	No — manual rule update	**Yes — Evolution Intel**
India regulatory compliance	Partial — no STR/DPDP	**Yes — full compliance**

Why SentinelAI Wins

SentinelAI addresses the full scope of Problem Statement 2 — not just classification, but intelligence. By combining supervised ML with unsupervised anomaly detection, behavioural genome clustering, and Explainable AI, it delivers a complete fraud intelligence platform purpose-built for Bank of India.

Four capabilities no traditional rule-based system has: detecting brand-new fraud patterns it has never seen before, grouping accounts into fraud archetypes investigators can understand, explaining every single flag with feature-level attribution, and tracking how criminal strategies evolve over time.

Fraud Genome	Anomaly Detect	XAI Engine	India First
Criminal DNA profiling	Catches the never-seen	Legally defensible	DPDP + RBI + FIU-IND